Akoizo Original Practice - Form 2

Division C | Wright Stuff | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) If airspeed doubles with all else equal, how does lift change?

Unit: Wings and slow flight.
Worked-case data: A 9.5 g plane (with motor) has 0.040 m^2 of wing and C_L = 0.8. Find its level-flight speed (ρ = 1.2 kg/m^3).
Separate investigation observations:
Plane 1: Wing area 0.030 m^2; cruise 2.6 m/s; flight 85 s.
Plane 2: Wing area 0.042 m^2; cruise 2.1 m/s; flight 118 s.
Same: Mass, motor, and turns.

A. It halves

B. It stays the same

C. It becomes four times larger

D. It doubles

Answer: _____

2. (2 points) How many test flights does a complete log require?

Unit: Testing, the flight log, and competition day.
Worked-case data: Log data at a 9 m ceiling: torque 0.10 -> 95 s, 0.13 -> 118 s, 0.16 -> 104 s (ceiling bumps). Plan the first official flight.
Separate investigation observations:
Room A, 7 m ceiling: Best time at torque 0.10.
Room B, 11 m ceiling: Best time at torque 0.15.
Competition: Ceiling announced as 9 m.

A. 10 or more

B. 5

C. 20

D. 1 per motor

Answer: _____

3. (2 points) A glide shows repeated nose-up, nose-down stalls. What is likely?

A. Too much decalage or CG too far back

B. CG too far forward

C. Not enough dihedral

D. Propeller too small

Answer: _____

4. (2 points) A plane hits the ceiling early in the flight. What is the best first change?

A. Back off turns to reduce launch torque

B. Wind more turns

C. Use a thicker motor

D. Add camber

Answer: _____

5. (4 points) Why is a complete flight log worth building?

Unit: Rules, scoring, and the flight log bonus.
Worked-case data: A team with a complete flight log flies official flights of 96.4 s and 112.0 s. What is its score and tiebreaker?
Separate investigation observations:
Team A: Best flight 120 s, no flight log.
Team B: Best flight 105 s, complete flight log.
Team C: Best flight 112 s, partial flight log.

Answer: _____
_____
_____

6. (4 points) Why add ballast where it improves balance rather than anywhere?

Unit: Building light and strong.
Worked-case data: A built airframe weighs 7.4 g. The CG is slightly behind the target. How should the team meet the 8.0 g minimum?
Separate investigation observations:
Flight 1: Fresh build, 102 s.
Flight 6: After a week in a hot car, 61 s; plane turned sharply right.
Inspection: Right wing twisted 2 degrees.

Answer: _____
_____
_____

7. (4 points) How does propeller choice affect flight time directly?

Unit: Propellers: turning torque into thrust.
Worked-case data: A plane cruises at 1.2 m/s and its propeller effectively advances 0.24 m per turn. A motor has 1,200 usable turns. Estimate motor run time.
Separate investigation observations:
Prop 1: Diameter 22 cm; plane cruised level; motor ran 140 s.
Prop 2: Diameter 26 cm, same pitch; plane sank after the burst; 90 s.
Prop 3: Diameter 18 cm; plane climbed fast; motor ran 95 s.

Answer: _____
_____
_____

8. (4 points) How does camber help an indoor wing?

Unit: Wings and slow flight.
Worked-case data: A 9.5 g plane (with motor) has 0.040 m^2 of wing and C_L = 0.8. Find its level-flight speed (ρ = 1.2 kg/m^3).
Separate investigation observations:
Plane 1: Wing area 0.030 m^2; cruise 2.6 m/s; flight 85 s.
Plane 2: Wing area 0.042 m^2; cruise 2.1 m/s; flight 118 s.
Same: Mass, motor, and turns.

Answer: _____
_____
_____

9. (4 points) Name three useful extra log parameters.

Unit: Testing, the flight log, and competition day.
Worked-case data: Log data at a 9 m ceiling: torque 0.10 -> 95 s, 0.13 -> 118 s, 0.16 -> 104 s (ceiling bumps). Plan the first official flight.
Separate investigation observations:
Room A, 7 m ceiling: Best time at torque 0.10.
Room B, 11 m ceiling: Best time at torque 0.15.
Competition: Ceiling announced as 9 m.

Answer: _____
_____
_____

10. (4 points) Explain how dihedral rolls a plane level.

Unit: Stability, trim, and circling flight.
Worked-case data: A plane stalls on every glide test. List adjustments in order to try.
Separate investigation observations:
Glide: Smooth and straight.
Powered launch: Plane rolls left and spirals down in the first 5 s.
After 5 s: Torque drops; plane recovers if still airborne.

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: A team uses a 15:1 winder and tests show a motor breaks near 1,400 turns. How many cranks should they wind for a safe 8 percent back-off?

Answer: _____
_____
_____

12. (2 points) Which team scores highest?

Supplied synthetic observations:
Team A: Best flight 120 s, no flight log.
Team B: Best flight 105 s, complete flight log.
Team C: Best flight 112 s, partial flight log.

A. Team B

B. Team A

C. Team C

Answer: _____

13. (2 points) What caused the drop in performance?

Supplied synthetic observations:
Flight 1: Fresh build, 102 s.
Flight 6: After a week in a hot car, 61 s; plane turned sharply right.
Inspection: Right wing twisted 2 degrees.

A. The motor lost mass

B. The ceiling was lower

C. Heat warped the wing, changing trim

Answer: _____

14. (2 points) Define “Thrust” in the context of Wright Stuff.

Answer: _____
_____
_____

15. (2 points) Define “Lift coefficient (C_L)” in the context of Wright Stuff.

Answer: _____
_____
_____

16. (2 points) Define “Peak altitude” in the context of Wright Stuff.

Answer: _____
_____
_____

17. (2 points) Define “Decalage” in the context of Wright Stuff.

Answer: _____
_____
_____

18. (2 points) Define “Stretch-winding” in the context of Wright Stuff.

Answer: _____
_____
_____